08_fill_factor_fragmentation

Propietario	© Carlos Roberto
Etiquetas	

PURPOSE

- Analyze index fragmentation using sys.dm_db_index_physical_stats.
- Determine when to use REBUILD vs REORGANIZE based on fragmentation thresholds.
- Document the fill factor decision for STN based on insert and update patterns.

8.0 ENVIRONMENT CHECK

```
USE STN_Lab;
GO

-- ============================================================
==
-- 8.0 -- ENVIRONMENT CHECK
-- Fill factor actual de cada indice en STN.
-- ============================================================
==

SELECT
    t.name                          AS table_name,
    i.name                          AS index_name,
    i.type_desc,
    i.fill_factor,
    s.used_page_count * 8           AS size_kb,
    s.row_count
FROM sys.indexes i
JOIN sys.tables t
    ON i.object_id = t.object_id
```

```sql
JOIN sys.dm_db_partition_stats s
    ON i.object_id = s.object_id
   AND i.index_id  = s.index_id
WHERE t.name IN ('Contribuyente', 'Declaracion', 'Pago', 'Aud
itoriaFiscal')
  AND i.index_id > 0
ORDER BY t.name, i.index_id;
GO
```

Results | Messages

	table_name	index_name	type_desc	fill_factor	size_kb	row_count
1	AuditoriaFiscal	PK_Auditoria	NONCLUSTERED	0	912	50000
2	Contribuyente	PK_Contribuyente	CLUSTERED	0	10584	200000
3	Contribuyente	IX_Contribuyente_NIT	NONCLUSTERED	0	4312	200000
4	Declaracion	PK_Declaracion	NONCLUSTERED	0	27792	1550000
5	Declaracion	IX_Declaracion_ContribuyenteID	NONCLUSTERED	0	27856	1550000
6	Declaracion	IX_Declaracion_Contribuyente_Periodo	NONCLUSTERED	0	55640	1550000
7	Declaracion	IX_Declaracion_Periodo_Contribuyente	NONCLUSTERED	0	55696	1550000
8	Declaracion	IX_Declaracion_Estado_Fecha	NONCLUSTERED	0	55560	1550000
9	Declaracion	IX_Declaracion_Fecha_Estado	NONCLUSTERED	0	55672	1550000
10	Pago	PK_Pago	NONCLUSTERED	0	26272	1200000

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (52) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 10 rows

8.1 ANALISIS DE FRAGMENTACIÓN

avg_fragmentation_in_percent: porcentaje de paginas fuera de orden lógico vs orden físico.
page_count: numero de paginas del índice.

```sql
SELECT
    t.name                                      AS table_name,
    i.name                                      AS index_name,
    i.type_desc,
    s.avg_fragmentation_in_percent,
    s.page_count,
    s.avg_page_space_used_in_percent,
    CASE
        WHEN s.avg_fragmentation_in_percent < 5  THEN 'Sin ac
cion'
```

```sql
        WHEN s.avg_fragmentation_in_percent < 30 THEN 'REORGA
NIZE'
        ELSE 'REBUILD'
    END AS accion_recomendada
FROM sys.dm_db_index_physical_stats(
    DB_ID(), NULL, NULL, NULL, 'SAMPLED') s
JOIN sys.indexes i
    ON s.object_id = i.object_id
   AND s.index_id  = i.index_id
JOIN sys.tables t
    ON i.object_id = t.object_id
WHERE t.name IN ('Contribuyente', 'Declaracion', 'Pago', 'Aud
itoriaFiscal')
  AND s.page_count > 8
ORDER BY s.avg_fragmentation_in_percent DESC;
GO
```

Results | Messages

	table_name	index_name	type_desc	avg_fragmentation_in_percent	page_count	avg_page_space_used_in_percent	accion_recomendada
1	Contribuyente	IX_Contribuyente_NIT	NONCLUSTERED	1.33587786259542	524	99.0025698047937	Sin accion
2	Contribuyente	PK_Contribuyente	CLUSTERED	0.53557765876052	1307	99.1248949839387	Sin accion
3	Declaracion	PK_Declaracion	NONCLUSTERED	0.231481481481481	3456	99.7148134420559	Sin accion
4	Declaracion	IX_Declaracion_ContribuyenteID	NONCLUSTERED	0.231481481481481	3456	99.7148134420559	Sin accion
5	Pago	PK_Pago	NONCLUSTERED	0.214329454990814	3266	99.8428836174945	Sin accion
6	Declaracion	IX_Declaracion_Fecha_Estado	NONCLUSTERED	0.202195262853842	6924	99.541957005189	Sin accion
7	Declaracion	IX_Declaracion_Contribuyente_Periodo	NONCLUSTERED	0.101141453547175	6921	99.5851124289597	Sin accion
8	Declaracion	IX_Declaracion_Periodo_Contribuyente	NONCLUSTERED	0.101053847264328	6927	99.4988386459106	Sin accion
9	Declaracion	IX_Declaracion_Estado_Fecha	NONCLUSTERED	0	6920	99.5995058067705	Sin accion
10	Pago	NULL	HEAP	0	10000	99.5548554484804	Sin accion
11	Declaracion	NULL	HEAP	0	13800	99.6053619965407	Sin accion
12	AuditoriaFiscal	NULL	HEAP	0	317	99.3595873486533	Sin accion
13	AuditoriaFiscal	PK_Auditoria	NONCLUSTERED	0	112	99.2551890289103	Sin accion

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (52) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 13 rows

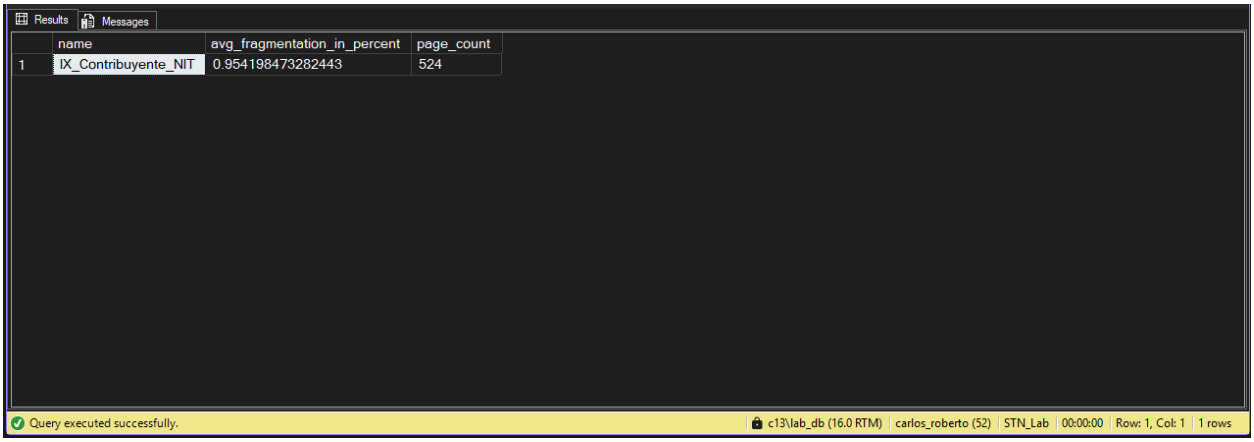
8.2 REORGANIZE vs REBUILD -- DECISION

REORGANIZE : operación online, desfragmenta nivel hoja. Usar cuando fragmentación entre 5% y 30%. No actualiza estadísticas.
REBUILD : recrea el índice completo. Puede ser offline. Usar cuando fragmentación supera 30%.
Actualiza estadísticas automáticamente.

8.2.1 Reorganice índice con fragmentación moderada

```sql
ALTER INDEX IX_Contribuyente_NIT ON dbo.Contribuyente REORGAN
IZE;
GO

-- Verificar fragmentacion despues del REORGANIZE
SELECT
    i.name,
    s.avg_fragmentation_in_percent,
    s.page_count
FROM sys.dm_db_index_physical_stats(
    DB_ID(), OBJECT_ID('dbo.Contribuyente'), NULL, NULL, 'SAM
PLED') s
JOIN sys.indexes i
    ON s.object_id = i.object_id
   AND s.index_id  = i.index_id
WHERE i.name = 'IX_Contribuyente_NIT';
GO
```

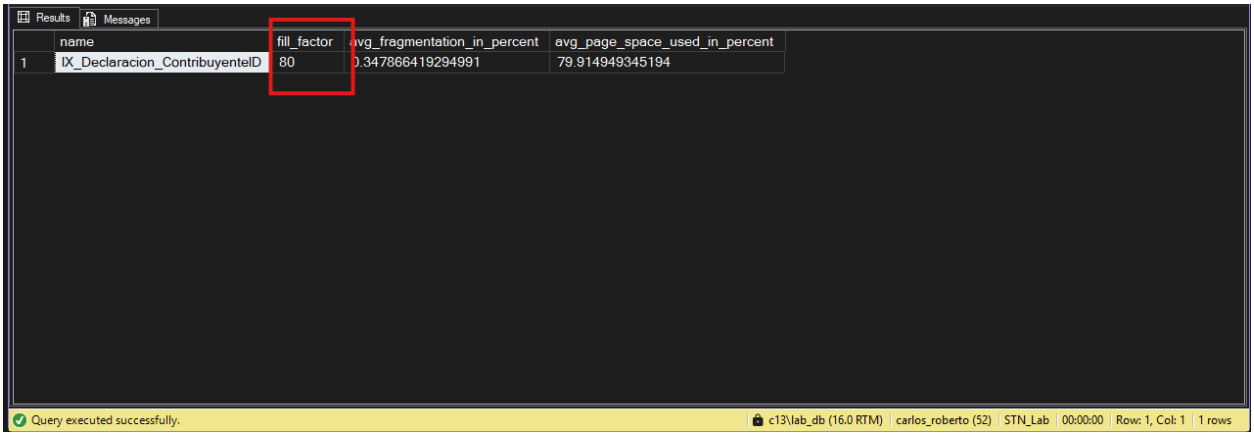


8.3 FILL FACTOR: IMPACTO EN FRAGMENTACIÓN

```sql
ALTER INDEX IX_Declaracion_ContribuyenteID
    ON dbo.Declaracion
    REBUILD WITH (FILLFACTOR = 80);
```

```sql
GO

SELECT
    i.name,
    i.fill_factor,
    s.avg_fragmentation_in_percent,
    s.avg_page_space_used_in_percent
FROM sys.dm_db_index_physical_stats(
    DB_ID(), OBJECT_ID('dbo.Declaracion'), NULL, NULL, 'SAMPL
ED') s
JOIN sys.indexes i
    ON s.object_id = i.object_id
   AND s.index_id  = i.index_id
WHERE i.name = 'IX_Declaracion_ContribuyenteID';
GO
```



Results | Messages

	name	fill_factor	avg_fragmentation_in_percent	avg_page_space_used_in_percent
1	IX_Declaracion_ContribuyenteID	80	0.347866419294991	79.914949345194

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (52) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 1 rows

8.4 SCRIPT DE MANTENIMIENTO AUTOMATIZADO

```sql
SELECT
    'ALTER INDEX [' + i.name + '] ON [dbo].[' + t.name + '] '
+
    CASE
        WHEN s.avg_fragmentation_in_percent < 5  THEN '-- Sin
accion requerida'
        WHEN s.avg_fragmentation_in_percent < 30 THEN 'REORGA
```

```sql
NIZE;'
        ELSE 'REBUILD WITH (FILLFACTOR = 80, ONLINE = OFF);'
    END AS maintenance_command,
    CAST(s.avg_fragmentation_in_percent AS DECIMAL(5,2)) AS f
ragmentation_pct,
    s.page_count
FROM sys.dm_db_index_physical_stats(
    DB_ID(), NULL, NULL, NULL, 'SAMPLED') s
JOIN sys.indexes i
    ON s.object_id = i.object_id
   AND s.index_id  = i.index_id
JOIN sys.tables t
    ON i.object_id = t.object_id
WHERE t.name IN ('Contribuyente', 'Declaracion', 'Pago', 'Aud
itoriaFiscal')
  AND s.page_count > 8
  AND i.index_id > 0
ORDER BY s.avg_fragmentation_in_percent DESC;
GO
```

Results | Messages

	maintenance_command	fragmentation_pct	page_count
1	ALTER INDEX [IX_Contribuyente_NIT] ON [dbo].[Contribuyente] -- Sin accion requerida	0.95	524
2	ALTER INDEX [PK_Contribuyente] ON [dbo].[Contribuyente] -- Sin accion requerida	0.54	1307
3	ALTER INDEX [IX_Declaracion_ContribuyenteID] ON [dbo].[Declaracion] -- Sin accion requerida	0.35	4312
4	ALTER INDEX [PK_Declaracion] ON [dbo].[Declaracion] -- Sin accion requerida	0.23	3456
5	ALTER INDEX [PK_Pago] ON [dbo].[Pago] -- Sin accion requerida	0.21	3266
6	ALTER INDEX [IX_Declaracion_Fecha_Estado] ON [dbo].[Declaracion] -- Sin accion requerida	0.20	6924
7	ALTER INDEX [IX_Declaracion_Contribuyente_Periodo] ON [dbo].[Declaracion] -- Sin accion requerida	0.10	6921
8	ALTER INDEX [IX_Declaracion_Periodo_Contribuyente] ON [dbo].[Declaracion] -- Sin accion requerida	0.10	6927
9	ALTER INDEX [IX_Declaracion_Estado_Fecha] ON [dbo].[Declaracion] -- Sin accion requerida	0.00	6920
10	ALTER INDEX [PK_Auditoria] ON [dbo].[AuditoriaFiscal] -- Sin accion requerida	0.00	112

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (52) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 10 rows